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Quantitative Time-Series Analysis of Sector ETFs:
XLK, XLE, and XLF

Dataset & Source

This project analyzed three U.S. sector exchange-traded funds (ETFs): XLK (Technology), XLE (Energy), and XLF (Financials). Historical daily price data beginning January 1, 2019 was collected using the Yahoo Finance API through Python's yfinance package.

These ETFs were selected to compare the behavior of different sectors across multiple market environments, including the COVID-19 crash, post-pandemic recovery, inflationary periods, and recent market trends.

Methods

The data was cleaned by extracting daily closing prices, sorting observations by date, and removing missing values. Daily percentage returns were then calculated to standardize comparisons across ETFs.

Several exploratory data analysis techniques were applied, including:

• Growth of a normalized $100 investment
• 30-day rolling annualized volatility
• Correlation heatmap of daily returns

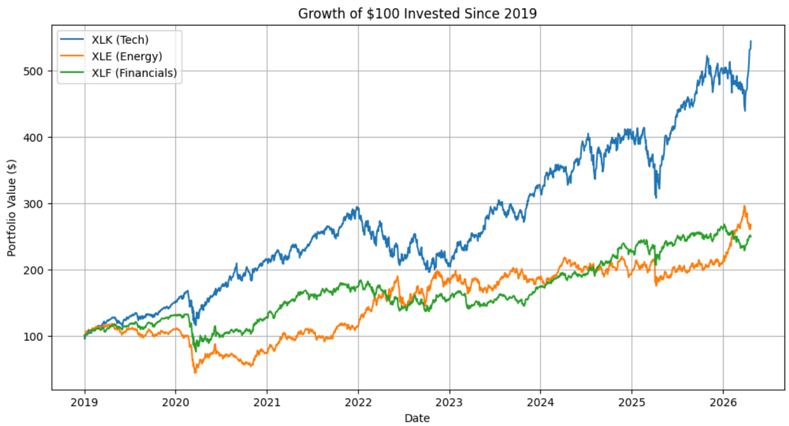
To satisfy the forecasting component of the project, Exponential Smoothing was applied to XLK price data. The final 60 trading days were reserved as a test set, and forecasting accuracy was evaluated using RMSE and MAE.

Key Findings

The Technology sector ETF (XLK) significantly outperformed the other sectors over the sample period, growing a hypothetical $100 investment to more than $500.

The Energy sector ETF (XLE) displayed the greatest volatility and strongest cyclical behavior, including a major drawdown during 2020 followed by a sharp recovery during inflationary and commodity-driven market conditions.

The Financial sector ETF (XLF) produced steadier but more moderate long-term growth.



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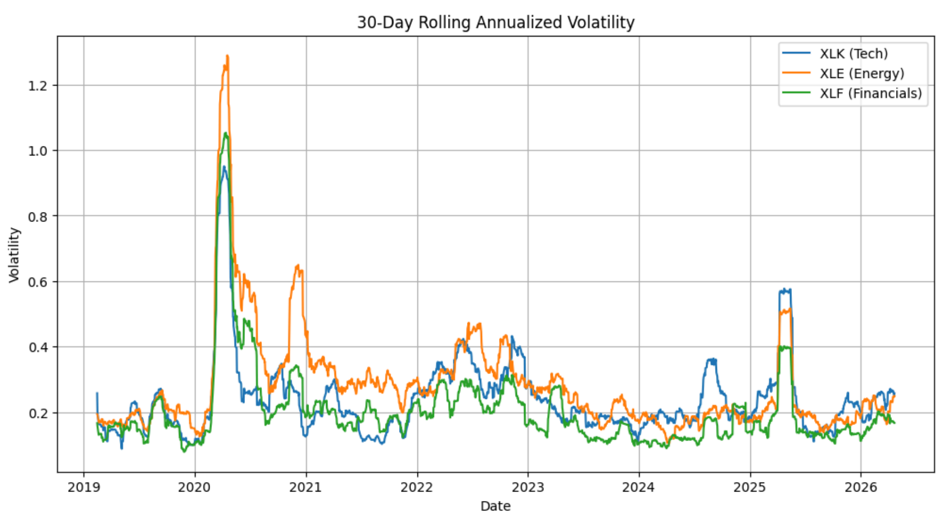
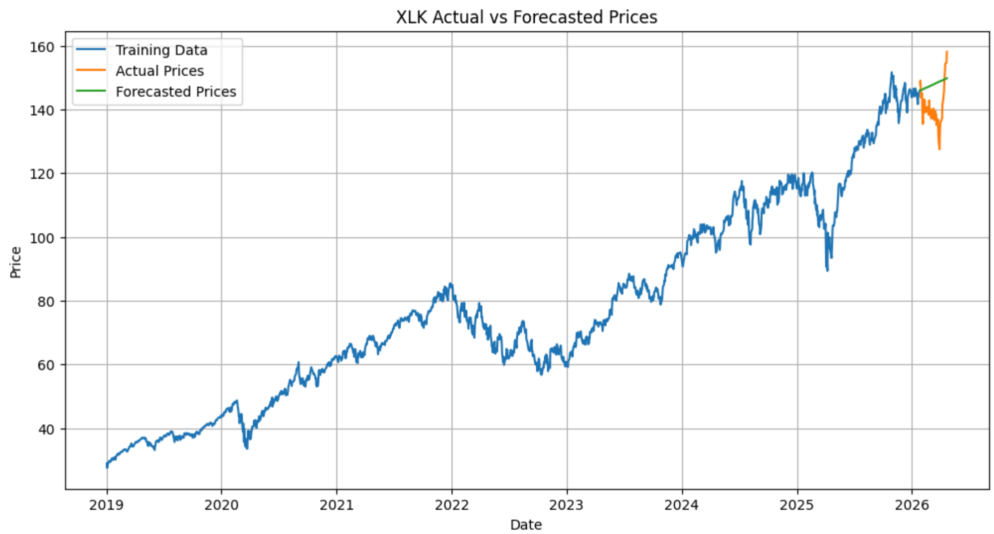
To better evaluate risk differences across sectors, rolling volatility was calculated using a 30-day annualized standard deviation of daily returns.



The rolling volatility chart highlights periods of market stress and changing sector risk over time. Volatility increased sharply during the 2020 COVID-19 market shock, with Energy showing the largest spike among the three ETFs.

Technology generally exhibited lower sustained volatility than Energy, while Financials remained moderate across much of the sample period. These results suggest that sector choice can significantly affect portfolio risk exposure.

To evaluate predictive performance, a time-series forecasting model was applied to XLK using Exponential Smoothing. The final 60 trading days were reserved as a test set and compared against model forecasts.



The forecasting model successfully captured the broader upward trend in XLK prices, but it was less effective at predicting short-term fluctuations and sudden market movements. This is common in financial markets, where prices are influenced by rapidly changing information and investor sentiment.

Overall, this project demonstrated that financial time-series data can be effectively cleaned, analyzed, and modeled using Python. Comparing sectors through returns, volatility, and forecasting provides deeper insight than observing raw prices alone.

The results suggest that Technology delivered the strongest long-term growth, Energy carried the highest cyclical risk, and Financials offered more moderate performance characteristics. While simple forecasting models are useful for identifying trends, precise short-term market prediction remains difficult.